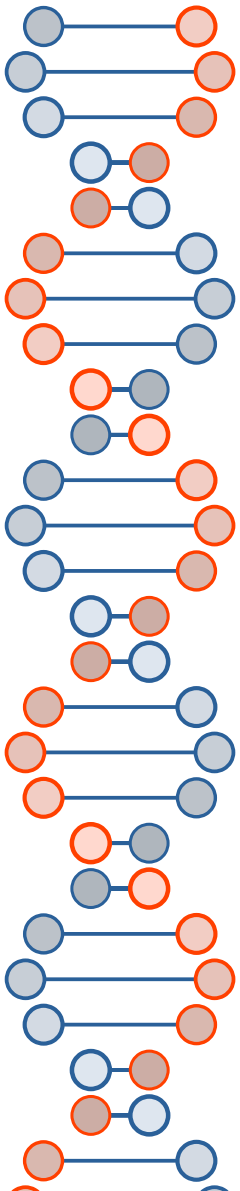


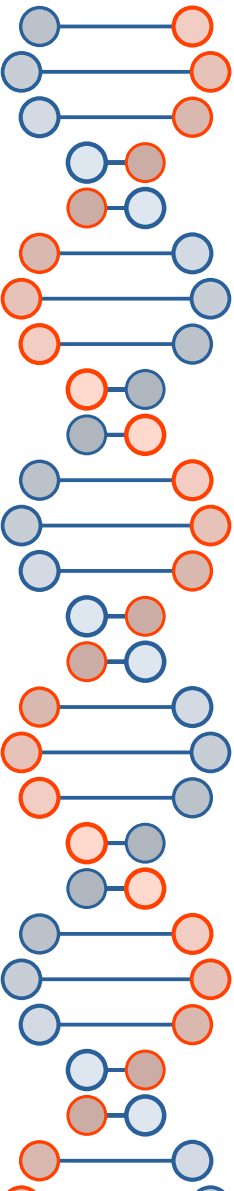
# Scaffolded DNA Computing

A brief description



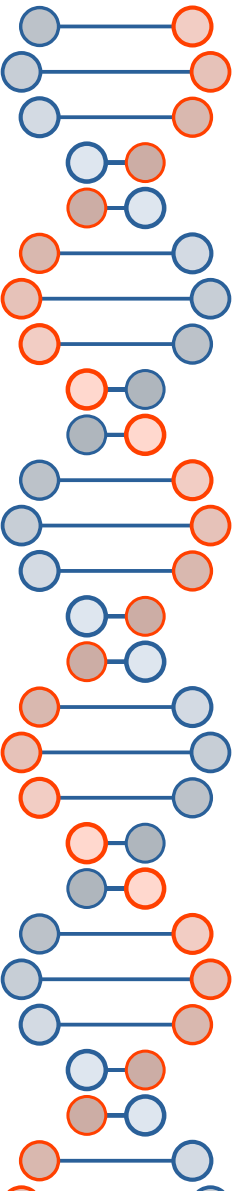
# Scaffolds and Tiles

- A scaffold is a strand of DNA upon which computations take place
- A tile is a unit of computation consisting of both the data and code sections of a program



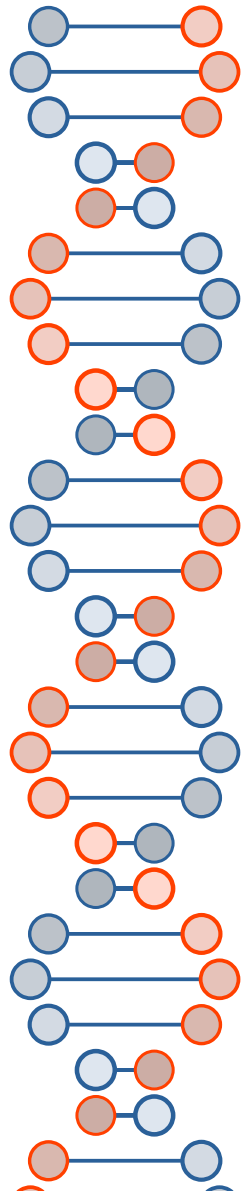
# Simulated Annealing versus Annealing

- Greedy algorithms are prone to finding themselves trapped within local minima
- Simulated annealing is an approach to counter this
- Start with a high temperature, and allow to cool
- At the higher temperatures solutions are generated that allow the process to move away from local minima
- As the temperature lowers it becomes less likely that a sub-optimal solution will be accepted



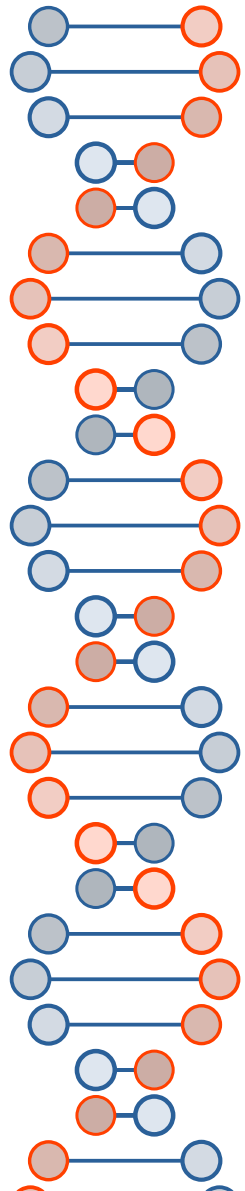
# Simulated Annealing versus Annealing

- With SDC actual annealing is a constituent part of the process
- The constituents are heated, and allowed to cool
- Solutions are reached that are “thermodynamically favoured”



# The Process

- $N + 1$  scaffold positions of which
- 1 position will be occupied by the anchor tile
- $N$  positions will be occupied by the compute tiles
- At the start of the process the tiles will assemble randomly according to binding energies
- As the temperature reduces the binding will favour configurations that make more sense thermodynamically



## Further Research

- Modelling the process using stochastic, or Monte Carlo, simulations
- Qualitative analysis of the process – understanding the effects of temperature, concentrations, and motion on the outcome (stoichiometric factors)
- Not your father's complexity analysis – a non-Euclidean theory of complexity?